ELNAZ HEIDARI

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EDUCATION

Ph.D. in Engineering, Clemson University, SC *2022 — 2026*
Dissertation: Explainable physics-guided Transformers for large-scale spatiotemporal flood forecasting.

M.Sc. in Computer Science, Georgia Institute of Technology, GA *2024 — 2026*
Specialization: artificial intelligence and machine learning.

RESEARCH AND PROJECT EXPERIENCE

Hydro-Transformers [Transfer Learning, Physics-guided DL, Spatiotemporal Forecasting, CUDA, PyTorch, Shell Scripting, HPC Cluster] Aug 2022 — Apr 2026

As a Graduate Research Assistant at Clemson University, developed a Temporal Fusion Transformer framework that forecasts river flow 1–24 hours ahead across 15 U.S. regions, reducing RMSE by 30% compared with NOAA's operational National Water Model. A single model learns from 12 weather time series and 27 catchment attributes, quantifies uncertainty through quantile regression, explains its predictions with attention, and generalizes zero-shot to 10 unseen regions.

Reinforcement Learning and Probabilistic State Estimation [PPO, Actor-Critic, GAE, MDPs, Q-Learning, SARSA, Kalman Filter, Particle Filter, PyTorch] Fall 2026

Implemented PPO from scratch in PyTorch, solving continuous-control LunarLander with a 223.7 mean return (threshold: 200) and running multi-seed ablations. Modeled traffic-signal control as an MDP, where model-based planning had 35% fewer congestion violations than Q-learning. Built Kalman and particle filters for multi-target tracking and satellite localization.

Generative Modeling on MNIST and Fashion-MNIST data utilizing VAEs, GANs and Diffusion Models [PyTorch, PyYAML, Matplotlib, CUDA] Spring 2025

Implemented and benchmarked VAEs, GANs, and diffusion models, analyzing the trade-offs between sample sharpness, diversity, and training stability. Diffusion models produced the sharpest and most stable samples, GANs risked mode collapse, and VAEs traded sharpness for diversity.

Distributional Semantics and Word Embeddings [PyTorch, GloVe, CBOW, Skip-Gram, Cosine Similarity] Fall 2025

Trained custom Word2Vec models (CBOW and Skip-Gram) on WikiText-2 (2M+ tokens), reaching up to 70% analogy accuracy and 0.74 cosine similarity on semantic relationships.

Language Modeling and Text Generation [PyTorch, NumPy, NLP, RNNs, LSTMs, Transformers, Attention Mechanisms, Gradient Clipping, Positional Encoding] Fall 2025

Built RNN, LSTM, and attention-based language models on a custom *Silmarillion* corpus, reducing test perplexity by about 25% through architecture and training improvements, and generated text with temperature-based sampling.

RNNs for Flood Forecasting [LSTM, GRU, Python, HPC Cluster] Aug 2022 — Dec 2025

Benchmarked LSTM, GRU, and vanilla RNN flood models against NOAA's National Water Model in 4 watersheds, using NOAA reanalysis weather and USGS gauge data (published in *Hydrology*, 2025).

Flood Mapping using Crowd-Sourced Data [Machine Learning, Decision Trees, ArcPy, JupyterHub, Python, GitHub] **Summer 2023**

As a Research Assistant at NOAA's National Water Center, built a decision-tree flood classifier that fuses Twitter/X posts and FEMA crowd-sourced reports with NWM inundation maps, improving flood-extent mapping over the NWM alone.

TECHNICAL SKILLS

Languages	Python, C/C++, Go, R, MATLAB, SQL, Bash
ML Frameworks	PyTorch, TensorFlow, Keras, scikit-learn, CUDA, NumPy, Pandas, SciPy, OpenCV
Deep Learning	Transformers, RNNs/LSTMs, CNNs, diffusion models, GANs, VAEs, attention
ML Methods	Time-series forecasting, uncertainty quantification, transfer learning, explainable AI
Reinforcement Learning	PPO, actor-critic, Q-learning, SARSA, value/policy iteration, MDPs
State Estimation	Kalman filters, particle filters, Bayesian filtering
Data & Infrastructure	HPC GPU clusters, AWS (S3, CLI), NetCDF4, Zarr, HDF5, Xarray, GRIB2, ArcPy, Git

SELECTED GRADUATE COURSES

Reinforcement Learning	Fall 2026	**AI for Robotics**	Fall 2026
Machine Learning	Spring 2026	**AI, Ethics, and Society**	Summer 2026
Natural Language Processing	Fall 2025	**Computer Networks**	Summer 2025
Deep Learning	Spring 2025	**Machine Learning for Trading**	Fall 2024
Data Structures and Algorithms	Spring 2024	**ML for Climate Adaptation**	Fall 2023

PUBLICATIONS

- **Heidari, E.**, and Khan, A. A. (2026). Probabilistic Sub-Daily Streamflow Forecasting in Data-Scarce Catchments Through a Zero-Shot Inductive Transfer Learning Approach. *Journal of the American Water Resources Association (JAWRA).*
- **Heidari, E.**, and Khan, A. A. (2026). Interpretable Hourly Flood Forecasting Leveraging Deep Learning Recurrence and Attention Mechanisms. *Journal of Water Resources Planning and Management* (accepted).
- **Heidari, E.**, Samadi, V., and Khan, A. A. (2025). Leveraging Recurrent Neural Networks for Flood Prediction and Assessment. *Hydrology*, 12(4), 90.
- **Heidari, E.**, Samadi, V., and Popescu, I. (2023). Flood Modeling Across Mountainous and Coastal Watersheds: Coupling Data-Driven and Catchment Physical Attributes. *AGU Fall Meeting.*
- **Heidari, E.**, and Samadi, V. (2023). Application of Deep Recurrent Neural Networks for Flood Prediction and Assessment. *EGU General Assembly*, EGU-16678.